

Pre- feasibility study to establish Gypsum factory



2017



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1. Exclusive summary

The project aims to exploit the gypsum ore in Tafila, and establish a factory manufacturing of all kinds of gypsum, and used in many industries, such as chalk, construction, etc., in addition to the export gypsum abroad.

Table 1 Project Main Indicators

Indicator	
Total Investment Cost (Jordan Dinars)	2.043.567
Internal Rate of Return (IRR)	%29.99
Pay-Back Period (Years)	4
Net Present Value (Jordan Dinars)	3.139.783
Job Opportunities provided by the Project	14
Total Revenues (For 10 Years)	28.659.698
Total Operating Expenses (For 10 years)	20.334.056
Total Net Profit (For 10 years)	3.713.646

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

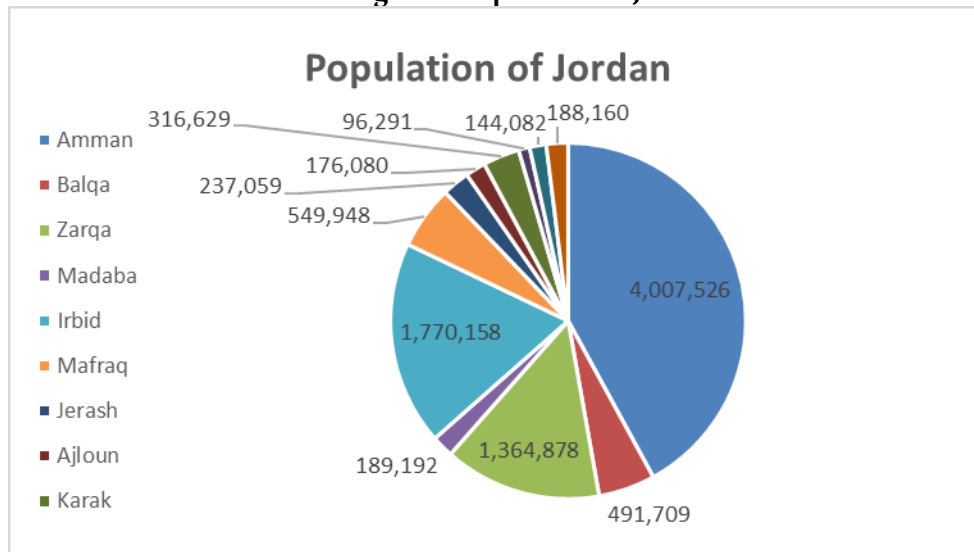
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2: Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

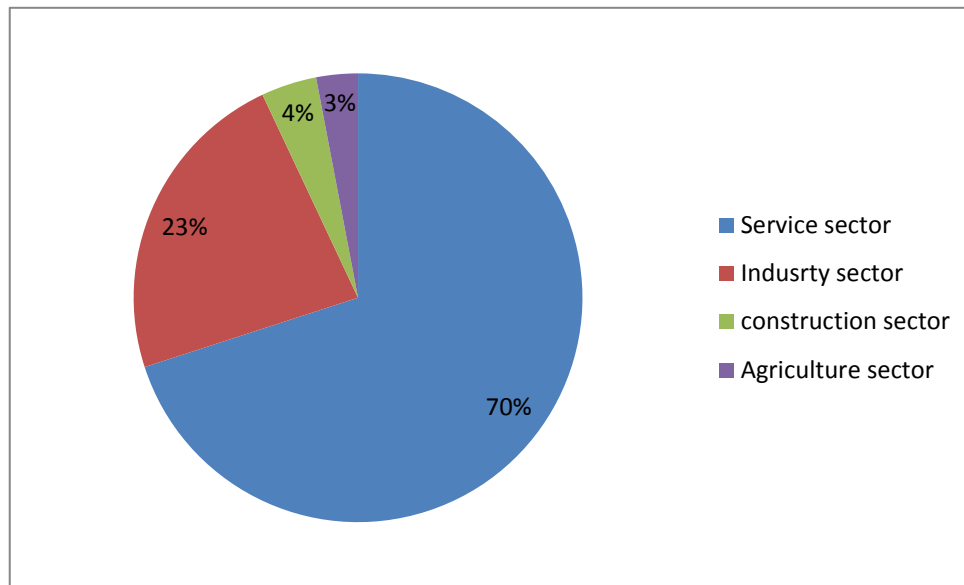
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in

licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5%

income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Geographic and demographic reality

Tafeileh

Jordan is one of the provinces which are located in the south of the kingdom, where one of the most important sanctuaries Middle East there is a Dana Biosphere Reserve, and away from the capital, Amman, about 180 km. The Tafeileh of the oldest populated areas, where successively by different nations: Starting Balodumien then the area subject to the rule of the Nabataeans that the Romans came to the area and while its former name was de Twlos means or chrome then underwent Islamic rule after the Battle of Mutah and the Battle of Yarmuk.

- Maintain Area: (2209.5) 2 km.
- Population density of the province (43.6) inhabitants / km².
- The unemployment rate for the year 2016: (17%).

Demographic studies Jordanian national census of 2004 indicates that Tafila had a population (81,000), but in 2010 about (85,600) people, and in 2016 the total population (96, 291) Source (Department of Statistics)

The investment environment

Tafila regarded with great natural resources but has not been exploited fully and the most important minerals that exist within it: Copper and estimated reserves of pure metal known about a million metric tons. The manganese, known reserves of it is estimated at about half a million metric tons. Tafila is rich in minerals and industrial rocks, which encouraged the Jordanian companies to exploit some of these resources, such as the Jordan Phosphate Mines Company, which makes phosphate ores in Hasa region, and Jordan Cement Factories Company for the production of cement in Rashadiya factory.

The most prominent investment opportunities in Tafila:

- exploit uranium to produce atomic energy
- Building Materials Industries (gypsum, zeolite)
- fertilizer industry and the chemical industry
- exploitation of oil shale to generate electricity

Ease of commercial Business

Jordanian Corporation for Enterprise Development works on the development of economic projects in the Kingdom and care, support and increase exports and export opportunities to the international and regional market size. And through the empowerment of Jordanian companies and projects of development and growth locally, and promote their products in the international markets

Project description

Ore gypsum has a great economic importance and many advantages since its used in multiple industries such as cement, building materials and other industrial aspect.

Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) where hydro calcium sulfate comprising 97.1%, as well as 2.9% water, in addition to other impurities such as calcium carbonate, silica sand of the total compound. As a non-metallic mineral sedimentary dominated by the white color when it is pure, but wight been exist in colors when accompanced with impurities. The gypsum deposited naturally as a result of the evaporation-rich fluid ions, calcium ions sulfates and exists in nature in many forms, such as gypsum crystal and needle and mass. Gypsum used in many industries, the most important, cement, ceramics, buildings, chemical and, medical industries.

Table 1 deposit areas gypsum

Areas	Formation
Central and southern parts of Jordan (Melih / Tafileh, Wadi Al Karak, Wadi Hasa, Wadi Mujib)	Fhais, homar, Shoaib, Wadi Seer
Alazraq	alazraq
Aldahel	Fhais, homar, Shoaib
Jabal Bani Hamida	shoaib
Alzrqaa River Basin / Subaihi	Abu-Royce

Source: Regulatory Authority of Energy and Mining sector.

Characteristics and advantages of gypsum:

Gypsum material has the characteristics and features makes it always as a basic material in the building materials industry , and most important of these features includes: fire resistance, absorption and sound insulation, heat insulation, mechanical properties good ranging of bending strength between 40-60 kg / cm², according to gypsum type used, and the percentage of water, it can also improve these properties, especially to increase hardness of the surface, giving a degree of good purity of different surfaces, beautiful white colour can be painted in any colour of paints, and stay for a long time, especially if used in method, easy to use and ability of composition in minutes, since its quickly getting hard.

Table 2 Chemical properties of gypsum

Area	SO ₄ %	CaO%
Central and southern parts of Jordan (Melih / Tafileh, Wadi Al Karak, Wadi Hasa, Wadi Mujib)	37-46	25-35
Alazraq	42-45	32-35
Aldahel	36-55	29-39
Jabal Bani Hamida	45-60	32-36
Alzrqaa River Basin / Subaihi	40-47	31-34

Source: Regulatory Authority of Energy and Mining sector.

• Natural gypsum

Natural gypsum is existing in rock salt in the form of flat objects or fibrous with limestone or sandstone or mud or in the form of deposits with thick layers in a wide extension of long as lenticular bodies- form of mono-crystals. Plaster and natural gypsum occurs several types as follows:

- Jbsat: non pure deposit of fine granules mixed with sand and clay.
- Fiber parallel: It is a lot of cracking plaster blocks exist in the form of a variety of fibers, characterized by pearly gloss.
- Plaster Rock: type a coherent crusty or grainy and often is not pure.

- Saliniet: one of the finest types of gypsum which consists of a full and transparent cracking a few single crystals.

Extraction method of gypsum

The mining of gypsum is carried out by miners exposed system (open cast) in places where the deposited materials is out cropped near the earth's surface, where started by removal of the overburden work benches in the form of horizontal layers starting from the top to the bottom in the form of a stairs. In case of thick sedimentary cover areas the underground mining system (Underground mining) will be used.

Aims of the project

- The approach of manufacturing the jordanian gypsum ores in a proper capacity comply with local marketed requirements by establishing a factory to produces gypsum in Tafieleh governorate, because of the quantities imported from abroad, which is about 36,773 ton per year.
- Encouraging the investors to be engaged in industries that rely on raw gypsum as a result of the rapid development witnessed by the world based on the industrial rocks, because of multiple industrial uses, which is available in quantities and good qualities in Jordan is one of the most important of these industrial rock.
- High possibility of Export of gypsum, whether as manufactured or as ore.

Gypsum uses

Gypsum has many industrial uses such as:

- cement industry which added by (3-6%) in order to delay the speed of intransigence, the glass industry to facilitate the degassing process, gum industry, is also used as filler in the paper industry, and in the installation of the mud used in oil drilling operations, and lime industry.
- used in construction and decoration work.
- chalk blackboards industry in educational and other institutions.

Gypsum enters in many industries that are affecting the daily lives of the human, and in different areas as follows:

- calcined gypsum (Jesse Paris) constitutes about 95% of the uses of gypsum, and purity ranging between 85-95%, depending mainly on gypsum heating to 130 ° C, therefore, losing water entering a weak link in

the composition transformed into plaster half-aqueous ($\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$) known commercially as Stucco, the most uses in construction, such as adhesives walls, panels adhesives, suspended ceilings, and lining the walls, and partitions.

- a gricultural gypsum uses to improve the properties of the soil to compensate the lack of soil in calcium, sulfur, and to address harmful salts and alkali in the soil, because it is characterized as uncostly, and easy to use, as well as, less its negative effects on the environment when used in comparison with other chemical additives used to improve the soil
- The excellent qualities of calcined gypsum is used in medical purposes to work splints bone fractures.

Targeted categories

Raw gypsum is used in many industries the following are the names of the mining companies with right license:

- Lafarj factory
- Hussin and ali almanaseer
- Jordan Cement Factories Company
- Rawabi Mining Company
- General Mining Company
- Export prospects

Analysis of markets

Gypsum reserves in Jordan is estimated, as shown in Table 3:

Table3 gypsum reserves

Area	Geological reserves(million ton)	Thickness
Alzrqaa River Basin / Subaihi	10	60
Central and southern part of jordan	8	0.2-4
Alazraq	3	0.25-0.5
Aldahel	0.284	1-3

Source: Regulatory Authority of Energy and Mining sector.

Competitors analysis

- General Mining Company

The General Mining Company mining gypsum ore from alzrqa (Subaihi River Basin mines) and the Melih and Alberbith (Tafileh) to meet established the needs of the cement Jordanian and factories private sector. The Arab company was established to manufacture gypsum with a capital of 12 million dinars, and began the process of production with the beginning of 2001

- Jordan Cement Factories Company
- Abbas Mining Corporation
- Shaker Taleb Foundation for Mining

It is recommended to concentrate on producing (a-Quality Plaster) which is most expensive since there is an opportunity for the export of this type to the Arab and international markets if they are produced according to international specifications.

Market share

Prices range in the domestic market is (50-70) Jordanian dinars per ton.

Based on a huge amounts of imports, estimated at 36,773 tons / year according to the Department of Statistics (2015) and the amount of domestic production estimated at 109,564 tons / year, the establishment of a factory to produce 50,000 tons plant / year, equivalent to 167 tons / day duration work 8 hours per day will contribute in reducing the import of this raw material and rely on domestic production

Price analysis and pricing policy

Annual production of these companies at estimated (13,465.2 tonnes) / year, 25,000 tons of gypsum were exported in 2004 as stated in the publications of the organization of the energy and minerals sector commission,

Table4 prices assumptions

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Sales price/ ton Gypsum	50	52	53	55	56	58	60	61	63	65

Estimated revenues

Table5 estimated revenues during the years 2018-2024

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Sale price/ ton Gypsum	50	52	53	55	56	58	60	61	63	65
Revenues	2500000	2575000	2652250	2731818	2813772	2898185	2985131	3074685	3166925	3261933

2. Technical analysis

Human resources required

Table 6 Human resources required for project

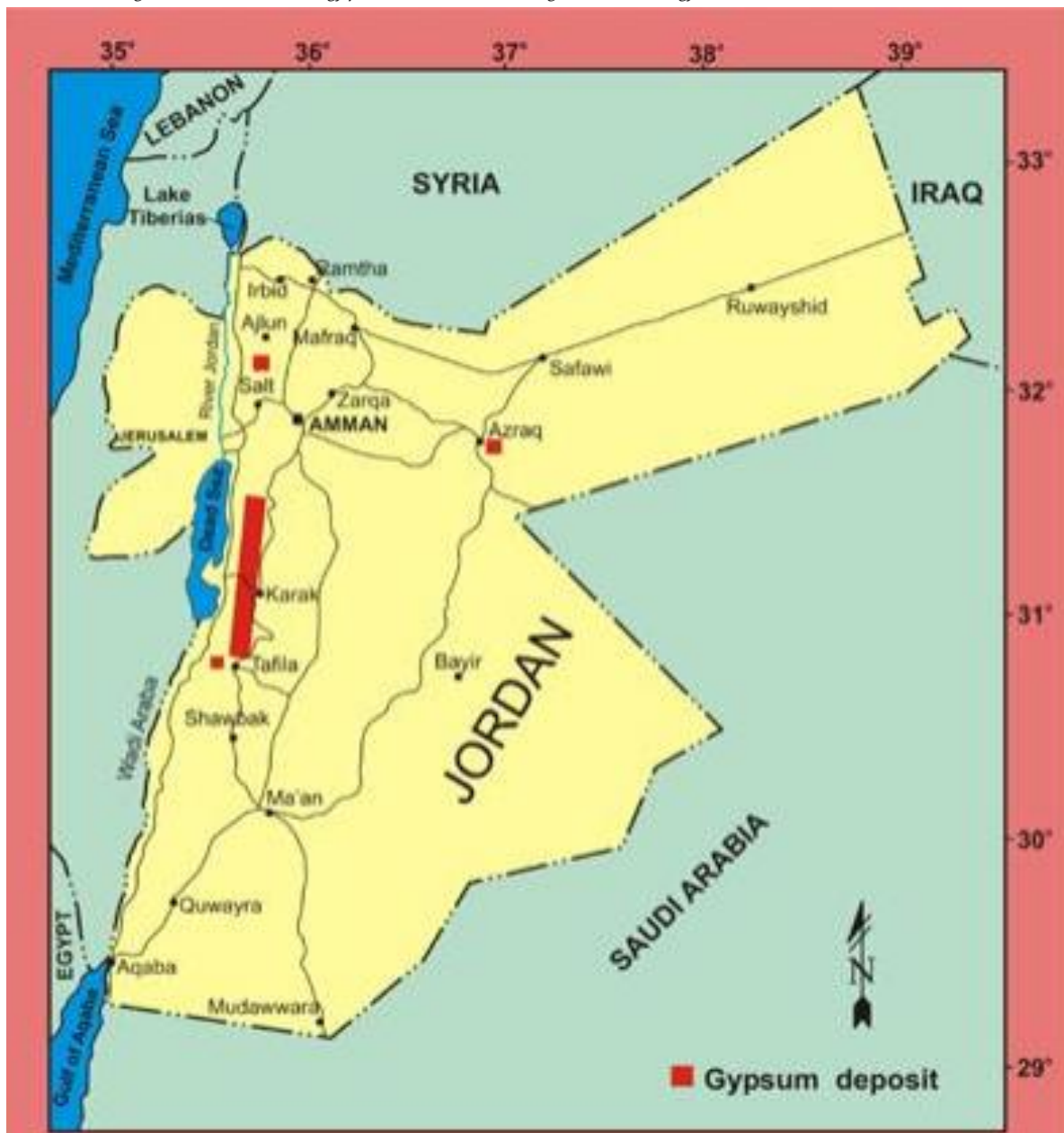
Job title	No.
Project manager	1
Geologist	1
Accountant	1
Technician	3
maintenance	1
Cleaning labour	1
driver	4
Security guard	2
Total	14

- Project Manager: supervising of work execution the proper of the staff, and take decisions related to the production process and the process of distribution and deployment of the product in the local market
- Geologist: apply the theoretical knowledge and research data available to determine the best locations of resources and potential future development sites
- Accountant: Follow up money matters and financial audit.
- Maintenance: Apply the procedures and activities of the electrical and mechanical maintenance, and follow up the implementation of plans and programs planned in order to ensure the continuity of the production process and quality at full capacity and low expenses as possible, and take care of providing spare parts required, and preparation of technical reports.
- Cleaning labour: to maintain the plant and offices clean.
- Driver: transfer the orders to customers.

Location analysis

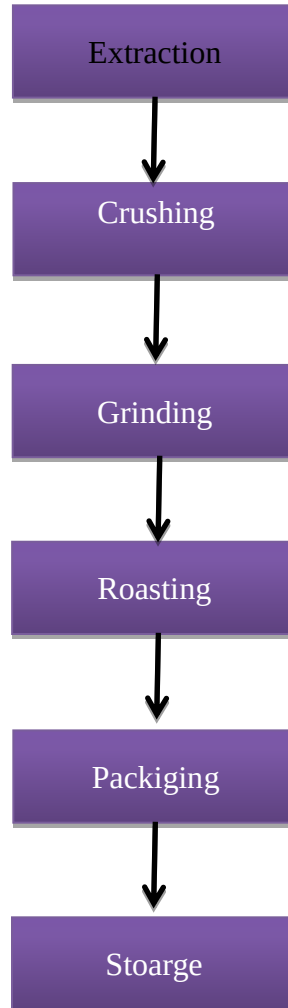
Since Melih area contains gypsum ore reserves estimated at 712,000 tons, with a thickness of the gypsum layer 1.5-2.5 m, this area is selected which is located about 30 km north of Tafilah with an area of 48 km².

Image (1): Distribution of gypsum ores in Jordan-regulate the energy and minerals sector commission.



Description of production process and steps of manufacturing

natural gypsum industry passes several stage as follow:



- crushing: break down and are extracted ore by crushing into small pieces in two phases, one initial cracking to reduce its size into small pieces the size of the palm of the hand, and the other secondary cracking up to the size of the lenses. And then stored in warehouses in preparation for sending it to the toaster.
- Grinding: gypsum after roasted will be send to mills for grinding, and can be calibrated these mills to obtain the required smoothness.
- Roasting: is sent gypsum crushed after extraction from the storage depots to private kilns process at a temperature of 130 ° C for roasted heat, stay inside long enough to expel three-quarters of the water that contains it raw gypsum, this results in two types of gypsum, namely: gypsm alfa half hydro and

Gypsum beta half hydro, both types are similar in crystalization, but the first is less likely to interact and thawing, therefore requires a large amount of water and a longer period of hardening, which is widely produced and used.

- Packing: the ground gypsum sends to private warehouses in preparation for packaging in bags. Pre-packaged in bags samples are taken of it to conduct a number of tests to determine the duration sclerosis, purity, type the crushing strength, bending, the type of impurities and the proportion of each type to be classified in the light of those results

Technical requirement analysis

Land and construction works

It was assumed that the factory will be established on a plot of land of 20 donums. The construction works required for the project is represented as follows

Table7 construction building

construction building	Cost(JD/M ²)	Area(M ²)
Building purposes	200	200
Hunger	200	1,000
paving	15	500
Other	-	-

Equipments and machines

The table (8) below is description the equipments and machines for the project:

Table8 the equipments and machines for the project

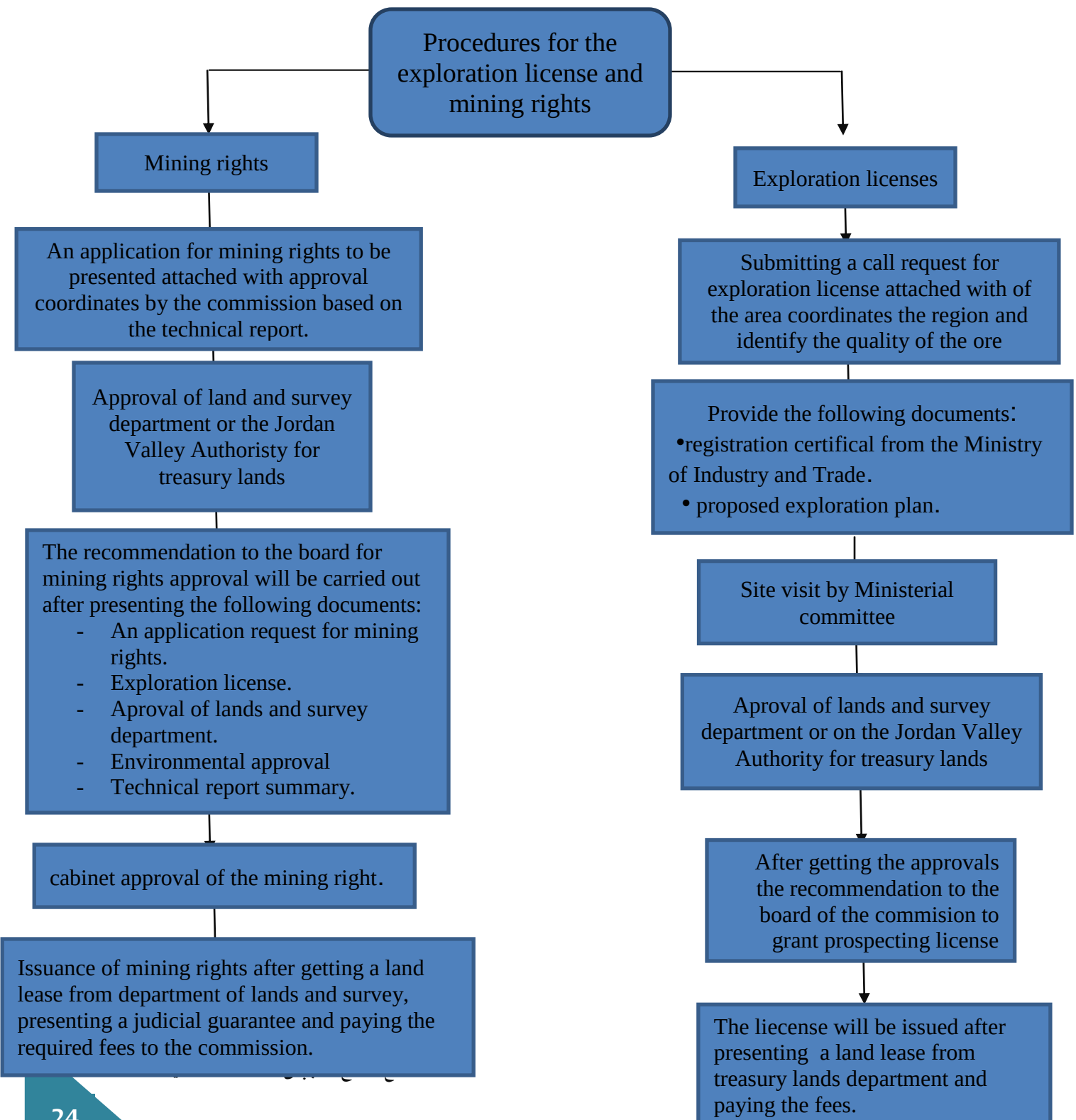
Machine	No.	Cost
Loder	2	60,000
Small dumber(12 ton)	2	100,000
Jawcusher	1	40,000
Cone crusher	1	70,000
Roasting machine	2	120,000
Seiving machine	1	25,000

Furnitures

The required furnitures for the project will be as follows: Office furnitures (computers, tables and chaires, carpet, meeting requierments) the cost 12,000 JD

Requirments and legal procedures

The following figure shows the necessary procedures for exploration and mining rights license based on regulary of energy and minerals sectors commision:



3. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is %10.75 as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 9 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table10 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 11 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table12 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	60.00%
Utilities Expenses from Total Revenues	2.5%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.25%

Table13 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
(2) Income Tax Deduction	0.50
Income Tax after Deduction	20%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	20.00%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table14 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%20
Bets	0.89
Equity	817,427
Loans Value	1,226,140
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	1,226,140
Loans Percentage	%60

Equity	
Cost of Equity	%15.5
Equity Value	817,427
Equity Percentage	%40
Gross	
Project Value	2,043,567
Weighted Average Cost of Capital	%10.75

Capital Expenditures

The estimated cost of the project is 2,043,567 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 15 Construction Work Capital Expenditure

Description	Cost (JOD per Squared)	Area	Gross Cost
Construction Works	200	200	40,000
Hungers works	200	1,000	200,000
Paving	15	500	7,500
Other Works			24,750
Total			272,250

Table16 machinery and equipments Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
Loder	2	60,000	120,000
Small dumber(12 ton)	2	100,000	200,000
Jawcusher	1	40,000	40,000
Cone crusher	1	70,000	70,000
Roasting machine	2	120,000	240,000
Seiving machine	1	25,000	25,000
Total			695,000

Table 17 Pre Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	500
Transportation	750
Marketing	500
Training	750
Total	2,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 18 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	443,438
General and administrative expenses	3	%25	5,838
Marketing expenses	3	%25	1,563
Indirect Salaries	3	%25	25,200
Total			476,038

Table19 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Construction works	272,250	10%	299,475	%14.7
Machinery & Equipment	695,000	10%	764,500	%37.4
Furniture	12,000	10%	13,200	%0.6
Vehicles	400,000	10%	440,000	%21.5
Pre-Operating Expenses	2,500	10%	2,750	%0.1
Working capital	476,038	10%	523,642	%25.6
Total (JD)	1,857,788		2,043,567	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (2,043,567) JD, where 60% of the project will be financed through loans (1,226,140) JD, and the remaining (40%) through (Equity) with (817,427) JD. The following table summarizes the general structure of the required financing:

Table 20 Sources of Funding

Funding Sources	Amount	Percentage
Equity	817,427	%40
Loans	1,226,140	%60
Total	2,043,567	%100
Use of Fund:	Value	Ratio
Capital Expenditures	1,517,175	%74.24
Pre-Operating Expenses	2,750	%0.13
Working Capital	523,642	%25.62
Total	2,043,567	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 21 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	1,500,000	1,545,000	1,591,350	1,639,091	1,688,263	1,738,911	1,791,078	1,844,811	1,900,155	1,957,160
Utilities from Revenues	62,500	64,375	66,306	68,295	70,344	72,455	74,628	76,867	79,173	81,548
Maintenance from Revenues	25,000	25,750	26,523	27,318	28,138	28,982	29,851	30,747	31,669	32,619
Disposables from Revenues	25,000	25,750	26,523	27,318	28,138	28,982	29,851	30,747	31,669	32,619
Others	161,250	166,088	171,070	176,202	181,488	186,933	192,541	198,317	204,267	210,395
Total	1,773,750	1,826,963	1,881,771	1,938,225	1,996,371	2,056,262	2,117,950	2,181,489	2,246,933	2,314,341

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table22 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	600	618	637	656	675	696	716	738	760	783
Stationary and Printing	600	618	637	656	675	696	716	738	760	783
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Rent	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Training	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Insurance	11,379	11,720	12,072	12,434	12,807	13,191	13,587	13,995	14,414	14,847
Miscellaneous	2,123	2,187	2,252	2,320	2,389	2,461	2,535	2,611	2,689	2,770
Total	23,352	24,052	24,774	25,517	26,283	27,071	27,883	28,720	29,581	30,469

Marketing Expenses

Table 23 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	6,250	6,438	6,631	6,830	7,034	7,245	7,463	7,687	7,917	8,155

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table24 Expected salary and wages

Job title	No.	Monthly salary (JD)	Total monthly salaries	The total annual salaries (JD)
Project manager	1	1,500	1,500	24,000
Geological man	1	550	550	8,800
acconter	1	400	400	6,400
Technical mining	3	350	1,050	16,800
maintenance	1	350	350	5,600
Cleaning man	1	250	250	4,000
driver	4	400	1,600	25,600
Security guard	2	300	600	9,600
Total	14			100,800

Table 25 Expected monthly salary and wages for the years (2018-2027)

Job description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Geological man	550	573	596	620	646	672	700	729	759	790
acconter	400	416	433	451	470	489	509	530	552	574
Technical mining	350	364	379	395	411	428	445	464	483	502
maintenance	350	364	379	395	411	428	445	464	483	502
Cleaning man	250	260	271	282	294	306	318	331	345	359
driver	400	416	433	451	470	489	509	530	552	574
Security guard	300	312	325	338	352	367	382	397	414	431

Table 26 Expected Annual Salaries and Wges (2027-2018)

Job description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Geological man	8,800	9,161	9,536	9,927	10,334	10,758	11,199	11,658	12,136	12,634
acconter	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Technical mining	16,800	17,489	18,206	18,952	19,729	20,538	21,380	22,257	23,169	24,119
maintenance	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Cleaning man	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
driver	25,600	26,650	27,742	28,880	30,064	31,296	32,579	33,915	35,306	36,753
Security guard	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Total	100,800	104,933	109,235	113,714	118,376	123,229	128,282	133,541	139,017	144,716

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 27 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Construction Works	299,475	%5.0	%0.0
Machinaries	764,500	%10.0	%2.0
Furniture	13,200	%10.0	%5.0
Vehicles	440,000	%10.0	%0.0

Table 28 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	1,517,175	1,533,125	1,549,414	1,566,049	1,583,039	1,600,392	1,618,115	1,636,219	1,654,711	1,673,601
Total Depreciation Expenses	136,744	138,339	139,968	141,631	143,330	145,065	146,838	148,648	150,497	152,386
Total Accumulated Depreciation	136,744	275,083	415,050	556,681	700,011	845,077	991,915	1,140,563	1,291,060	1,443,446
Total Additions	0	15,950	16,289	16,635	16,990	17,353	17,724	18,103	18,492	18,890
Total Net Book Values	1,380,431	1,258,043	1,134,364	1,009,368	883,028	755,315	626,201	495,656	363,651	230,154

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table29 Loan Details

Loan Value	1,226,140
Annual Interest Rate	9.48%
Loans Period	5
Loan Starts at year:	2018
Annual Payment	319,167
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				1,226,140
2019	319,167	116,238	202,929	1,023,211
2020	319,167	97,000	222,167	801,044
2021	319,167	75,939	243,228	557,816
2022	319,167	52,881	266,286	291,530
2023	319,167	27,637	291,530	0

Income Statement

Table 30 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	2,500,000	2,575,000	2,652,250	2,731,818	2,813,772	2,898,185	2,985,131	3,074,685	3,166,925	3,261,933
Gross Operating Revenues	2,500,000	2,575,000	2,652,250	2,731,818	2,813,772	2,898,185	2,985,131	3,074,685	3,166,925	3,261,933
Operating Expenses	1,773,750	1,826,963	1,881,771	1,938,225	1,996,371	2,056,262	2,117,950	2,181,489	2,246,933	2,314,341
	-	-	-	-	-	-	-	-	-	-
Gross Operating Profit	726,250	748,038	770,479	793,593	817,401	841,923	867,180	893,196	919,992	947,592
<i>Gross Profit Percentage</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>
Salaries and Benefits (Indirect Staff)	-100,800	-104,933	-109,235	-113,714	-118,376	-123,229	-128,282	-133,541	-139,017	-144,716
General and Administraive Expenses	-23,352	-24,052	-24,774	-25,517	-26,283	-27,071	-27,883	-28,720	-29,581	-30,469
Markeing Expenses	-6,250	-6,438	-6,631	-6,830	-7,034	-7,245	-7,463	-7,687	-7,917	-8,155
Pre Operating Expenses	-2,750									
Gross Indirect Expenses	-133,152	-135,423	-140,639	-146,060	-151,693	-157,546	-163,628	-169,948	-176,515	-183,340
Income before Interest, Depreciation and Tax	593,098	612,615	629,839	647,533	665,708	684,377	703,553	723,248	743,477	764,252
Fixed Assets Depreciations	-136,744	-138,339	-139,968	-141,631	-143,330	-145,065	-146,838	-148,648	-150,497	-152,386

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income before Tax and Interests	456,355	474,276	489,872	505,902	522,378	539,312	556,715	574,600	592,979	611,866
Bank Interests	-116,238	-97,000	-75,939	-52,881	-27,637	0	0	0	0	0
Income Before Tax	340,116	377,276	413,933	453,021	494,741	539,312	556,715	574,600	592,979	611,866
Income Tax	-93,023	-100,455	-107,787	-115,604	-123,948	-132,862	-136,343	-139,920	-143,596	-147,373
Net Profit	247,093	276,821	306,146	337,416	370,793	406,449	420,372	434,680	449,383	464,492
<i>Net Profit Percentage</i>	<i>%10</i>	<i>%11</i>	<i>%12</i>	<i>%12</i>	<i>%13</i>	<i>%14</i>	<i>%14</i>	<i>%14</i>	<i>%14</i>	<i>%14</i>
Compulsory Reserves	-24,709	-27,682	-30,615	-33,742	-37,079	-40,645	-42,037	-43,468	-44,938	-46,449
Retained Earnings	222,384	471,522	747,054	1,050,729	1,384,442	1,750,246	2,128,581	2,519,793	2,924,238	3,342,281

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 31 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	247,093	276,821	306,146	337,416	370,793	406,449	420,372	434,680	449,383	464,492
Bank Interests	116,238	97,000	75,939	52,881	27,637	0	0	0	0	0
Depreciation	136,744	138,339	139,968	141,631	143,330	145,065	146,838	148,648	150,497	152,386
Total Operating Cashflows before Additions to Working Capital	500,075	512,160	522,053	531,929	541,760	551,515	567,210	583,328	599,881	616,879
Inventory (Increase/Decrease)	-125,000	-3,750	-3,863	-3,978	-4,098	-4,221	-4,347	-4,478	-4,612	-4,750
Accounts Receivables (Increase/Decrease)	45,625	1,369	1,410	1,452	1,496	1,541	1,587	1,634	1,683	1,734
Working Capital (Increase/Decrease)	-79,375	-2,381	-2,453	-2,526	-2,602	-2,680	-2,761	-2,843	-2,929	-3,016
Net Cashflows from Operating Activities	420,700	509,779	519,600	529,402	539,158	548,835	564,449	580,485	596,952	613,862
Cashflows from Investments Activities										

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Fixed Assets (Procurement)	1,517,175	-15,950	-16,289	-16,635	-16,990	-17,353	-17,724	-18,103	-18,492	-18,890
	-									
Net Cashflows from Investments Activities	1,517,175	-15,950	-16,289	-16,635	-16,990	-17,353	-17,724	-18,103	-18,492	-18,890
	-									
Cashflows from Financing Activities										
Capital	817,427									
Loan Amortization	-202,929	222,167	243,228	-266,286	-291,530	0	0	0	0	0
		-	-							
Bank Interest Rate	-116,238	-97,000	-75,939	-52,881	-27,637	0	0	0	0	0
Loans	1,226,140	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,724,400	319,167	319,167	-319,167	-319,167	0	0	0	0	0
		-	-							
Net (Increase/Decrease) in Cash	627,925	174,661	184,144	193,600	203,001	531,482	546,725	562,381	578,460	594,972
Cashflows at the Beginning of Period	0	627,925	802,586	986,730	1,180,330	1,383,331	1,914,813	2,461,538	3,023,919	3,602,380
Cashflows at the End of Period	627,925	802,586	986,730	1,180,330	1,383,331	1,914,813	2,461,538	3,023,919	3,602,380	4,197,352

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 32 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	627,925	802,586	986,730	1,180,330	1,383,331	1,914,813	2,461,538	3,023,919	3,602,380	4,197,352
Inventory	125,000	128,750	132,613	136,591	140,689	144,909	149,257	153,734	158,346	163,097
Total current Assets	752,925	931,336	1,119,343	1,316,921	1,524,019	2,059,722	2,610,795	3,177,654	3,760,726	4,360,449
Non Current Assets										
Fixed Assets (net)	1,380,431	1,258,043	1,134,364	1,009,368	883,028	755,315	626,201	495,656	363,651	230,154
Total Non Current Assets	1,380,431	1,258,043	1,134,364	1,009,368	883,028	755,315	626,201	495,656	363,651	230,154
Total Assets	2,133,356	2,189,379	2,253,706	2,326,289	2,407,047	2,815,037	3,236,995	3,673,310	4,124,377	4,590,603
Liabilities										
Current Liabilities										
Payables	45,625	46,994	48,404	49,856	51,351	52,892	54,479	56,113	57,796	59,530
Remaining amount of Loan	222,167	243,228	266,286	291,530	0	0	0	0	0	0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total current Liabilities	267,792	290,222	314,690	341,386	51,351	52,892	54,479	56,113	57,796	59,530
Non Current Liabilities										
Long Terms Loans	801,044	557,816	291,530	0	0	0	0	0	0	0
Total Long Term Liabilities	801,044	557,816	291,530	0	0	0	0	0	0	0
Total Liabilities	1,068,836	848,038	606,220	341,386	51,351	52,892	54,479	56,113	57,796	59,530
Owners Equity										
Shareholders Contributions	817,427	817,427	817,427	817,427	817,427	817,427	817,427	817,427	817,427	817,427
Statutory Reseve	24,709	52,391	83,006	116,748	153,827	194,472	236,509	279,977	324,915	371,365
Retained Profits	222,384	471,522	747,054	1,050,729	1,384,442	1,750,246	2,128,581	2,519,793	2,924,238	3,342,281
Total Equity	1,064,520	1,341,341	1,647,487	1,984,903	2,355,696	2,762,145	3,182,517	3,617,197	4,066,580	4,531,073
Total Liabilities and Equity	2,133,356	2,189,379	2,253,706	2,326,289	2,407,047	2,815,037	3,236,995	3,673,310	4,124,377	4,590,603

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 33 Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	-2,043,567	423,45	493,82	503,31	512,76	522,16	531,48	546,72	562,38	578,46	594,97	9,162,748
Discount Factor	1.00	0.74	0.66	0.60	0.54	0.49	0.44	0.40	0.36	0.33	0.29	0.29
Net Present Value for Cash	-2,043,567	311,69	328,21	302,03	277,83	255,45	234,76	218,05	202,52	188,08	174,67	2,690,011

Table below (34) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 34 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-2,043,567	423,450	493,829	503,31	512,76	522,168	531,482	546,725	562,381	578,460	594,972
Project Value	2,043,567	1,620,11	1,126,28	622,97	110,21	411,958	943,440	1,490,165	2,052,547	2,631,007	3,225,979
Payback Period (Year)	4	1	1	1	1	0	0	0	0	0	0

Table 35 Financial Analysis Results

Indicator	Value
Weighted Average Cost of Capital (WACC)	%10.75
Net Present Value for Cashflows	3,139,783
Payback Period	4
Internal Rate of Return	%29.99

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 36 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%29.99	4	10.75%
Revenues declined by 10%	%27.57	4	10.75%
Operating Expenses Increased by 10%	%22.23	5	10.75%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 3,139,753 Jordan Dinars considering that the project provides 14 Job Opportunities for the governorate residents.